

## INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit:

2. Subject Name:      Metamorphic Petrology    L-T-P:      3-0-0    Credits:

3. Pre-requisites: Introduction to Earth Sciences, Mineralogy, Optical Mineralogy Lab

4. Syllabus and reference books:

### **Syllabus:**

Introduction; Types and Controls of metamorphism; Concepts of Metamorphic Isograd, Metamorphic Facies, Facies Series and Paired Metamorphic belt; Metamorphism in Relation to Orogenic Processes; Heat Flow and Metamorphism, Steady State Geotherms, Thermal models for Contact and Regional metamorphism, Metamorphic P-T-t Paths; Thermodynamics of Homogeneous and Heterogeneous Systems; Graphical Representation of Mineral Assemblages; Nucleation and Growth of Metamorphic Minerals; Metamorphism of Pelites, Calcareous Sediments and Mafic Igneous Rocks along Different Field Gradients; Migmatites; Crustal Anatexis and Granulite Facies Metamorphism; Fluids in Metamorphism, Buffering versus Fluid Infiltration; Concepts of Petrogenetic Grids, P-T Pseudosections and Geothermobarometry; Element Diffusion and its Application in Diffusion Chronometry.

### **Reference Books:**

- (1) Metamorphic phase equilibria and pressure-temperature-time paths by Spear, F. S.; 1995; Monograph Mineralogical Society of America, pp. 799.
- (2) An Introduction to Metamorphic Petrology by Bruce Yardley & Clare Warren; 2021; 2<sup>nd</sup> Edition, Cambridge University Press.
- (3) Petrogenesis of Metamorphic Rocks by Kurt Bucher & Martin Frey; 2002; 7<sup>th</sup> Edition; Springer-Verlag Berlin Heidelberg.
- (4) Principles of Metamorphic Petrology by R.H. Vernon & G.L. Clarke; 2008; Cambridge University Press.
- (5) Principles of Igneous and Metamorphic Petrology by Anthony R. Philpotts & Jay, J. Ague. 3<sup>rd</sup> Edition; 2022; Oxford University Press.
- (6) An Introduction to Igneous and Metamorphic Petrology by John D. Winter. 2001; Pearson College Div.

### 5. Lecture-wise break-up:

| Sl. No.               | Topic                                                                                                                                                                                                                                        | No. of lectures |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1.                    | Metamorphic Introduction; Types and Controls of metamorphism; Classification of Metamorphism; Concepts of Metamorphic Isograd, Metamorphic Facies, Facies Series and Paired Metamorphic belt; Metamorphism in Relation to Orogenic Processes | 5 hrs           |
| 2.                    | Heat Flow Modelling of Metamorphic System; Concept of Steady State and Transient Geotherms; Thermal models for Contact and Regional metamorphism; Metamorphic P-T-t Paths                                                                    | 5 hrs           |
| 3.                    | Nucleation and Growth of Metamorphic Minerals                                                                                                                                                                                                | 2 hrs           |
| 4.                    | Thermodynamics of Homogeneous and Hetrogeneous Systems                                                                                                                                                                                       | 4 hrs           |
| 5.                    | Graphical Representation of Mineral Assemblages;                                                                                                                                                                                             | 1 hr            |
| 6.                    | Metamorphism of Pelites, Calcareous Sediments and Mafic Igneous Rocks along Different Field Gradients                                                                                                                                        | 8 hrs           |
| 7.                    | Migmatites: Definition and Classification; Crustal Anatexis and Granulite Facies Metamorphism                                                                                                                                                | 4 hrs           |
| 8.                    | Fluids in Metamorphism, Buffering versus Fluid Infiltration                                                                                                                                                                                  | 2 hrs           |
| 9.                    | Concepts of Petrogenetic Grids, P-T Pseudosections and Geothermobarometry                                                                                                                                                                    | 5 hrs           |
| 10.                   | Element Diffusion and its Application in Diffusion Chronometry.                                                                                                                                                                              | 4 hrs           |
| Total number of hours |                                                                                                                                                                                                                                              | 40 hrs          |